

Week 8 Lab 6 Report

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Question 1 (2 marks):

It means the message is a sequence of bytes, represented by integers between 0 and 255.

Question 2 (3 Marks):

The bytes are decoded first before printing onto the terminal.

```
def on_message(client, userdata, msg):
    print(msg.topic, msg.payload.decode('UTF-8'))
```

Question 3 (2 marks):

We load the model once as a global variable. This ensures the next time we need to use the model; we do not have to reload it.

```
with session.graph.as_default():
    set_session(session)
    model = load_model(MODEL_NAME)
```

Question 4 (2 marks):

We froze it to save time. The model has been trained on previously on a huge dataset with relatively good performance. So, we do not have to train the entire model again, and only train the last few layers so the model can specifically learn about flowers. No, it is not necessary since we are not training the model at this point.

Question 5 (2 marks):

The classify function from the previous section is used to obtain the probability score and index of the class with the highest score based on the prediction.

```
def classify(model: Model, image: np.ndarray) -> tuple[str, float, int]:
    with session.graph.as_default():
        set_session(session)
        result = model.predict(image)
        themax = np.argmax(result)

    return (tpl[themax], result[0][themax], themax)

def classify_flower(filename: Path, data: np.ndarray) -> dict:
    print("Start classifying")
    label, score, win = classify(model, data)
    # So that json can serialize
    score = float(score)
    win = int(win)
    print("Done.")
    return {"filename": filename, "prediction": classes[win], "score": score,
            "index": win}
```

Question 6 (2 marks):

In `main()`, the following code is added to select all files in the samples directory, and the image is sent.

```
for file in glob.glob(os.path.join('samples', '*')):  
    send_image(client, file)
```

On receiving a message from the server, the result will be printed out. For each `filename`, `prediction` contains the predicted flower class, and `score` contains the probability score of that classification.

```
def on_message(client, userdata, msg):  
    print("Received message from server.")  
    resp_dict = json.loads(msg.payload)  
    filename = resp_dict["filename"]  
    prediction = resp_dict["prediction"]  
    score = resp_dict["score"]  
    print(f"{filename = }, {prediction = }, {score = }")
```

Question 7 (2 marks):

Classification accuracy of $15/16 = 0.9375$ on the sample images.

```
filename = 'samples/daisy2.jpeg', prediction = 'daisy', score = 0.3836599290370941  
Received message from server.  
filename = 'samples/daisy3.jpeg', prediction = 'daisy', score = 0.40494632720947266  
Received message from server.  
filename = 'samples/dandelion1.jpg', prediction = 'dandelion', score = 0.8832752108573914  
Received message from server.  
filename = 'samples/dandelion2.jpeg', prediction = 'daisy', score = 0.5575123429298401  
Received message from server.  
filename = 'samples/dandelion3.jpg', prediction = 'dandelion', score = 0.48122426867485046  
Received message from server.  
filename = 'samples/rose1.jpg', prediction = 'roses', score = 0.9877322912216187  
Received message from server.  
filename = 'samples/rose2.jpeg', prediction = 'roses', score = 0.9959697723388672  
Received message from server.  
filename = 'samples/rose3.jpg', prediction = 'roses', score = 0.9958575367927551  
Received message from server.  
filename = 'samples/sunflower1.jpeg', prediction = 'sunflowers', score = 0.9964033365249634  
Received message from server.  
filename = 'samples/sunflower2.jpeg', prediction = 'sunflowers', score = 0.9042001962661743  
Received message from server.  
filename = 'samples/sunflower3.jpeg', prediction = 'sunflowers', score = 0.9739369750022888  
Received message from server.  
filename = 'samples/tulip.jpg', prediction = 'tulips', score = 0.9738032221794128  
Received message from server.  
filename = 'samples/tulip1.jpg', prediction = 'tulips', score = 0.8482610583305359  
Received message from server.  
filename = 'samples/tulip2.jpg', prediction = 'tulips', score = 0.9626380801200867  
Received message from server.  
filename = 'samples/tulip3.jpg', prediction = 'tulips', score = 0.9965517520904541  
Received message from server.  
filename = 'samples/tulips2.jpg', prediction = 'tulips', score = 0.9626380801200867
```